GOAT-Pro (MLB) model experiments: everything tried, and what is still untried

As of 2026-10-26. Every experiment ran on public MLB data (Statcast via the `savant_mart`, plus AAA Hawk-Eye). Details live in `pipeline/RESULTS_MLB.md` (section numbers below as §n); pass rules were written in `COLLABORATION.md` before each run; raw numbers are in `pipeline/results/`.

How to read the results.

- **Honest harness:** pitcher-grouped cross-fit. Pitchers are split into 4 folds by `crc32(pitcher) % 4`, and no pitcher is graded by a model that trained on him.
- **Baseline:** Stuff+ v2 on the honest harness.
  - Per-pitch fit .058 / .061 (2025 / 2026).
  - Year-to-year stability (YoY) .79 / .82 / .82.
  - Plate-height leak .233.
- **"Next r":** the correlation of the grade with the next season's outcome.
- **"Significant" / ∗:** the paired-pitcher bootstrap's 95% interval excludes zero.

A. Foundations: data and harness

1. **Feature and booster caching** (`train_mlb.py`)
   - **Why:** tests were slow.
   - **Strategy:** a parquet cache of the feature frame, keyed on the data plus the code; a cache of trained boosters.
   - **Result:** 70 s → 22 s per test, with identical output. Kept.
2. **Full 2026 season** (`pitches_all`)
   - **Why:** the old view stopped on 2026-08-07.
   - **Data:** day-files after that date.
   - **Result:** +202k regular-season pitches; 2026 grades r .98 vs the truncated run. Kept (owner decision).
3. **2026 plate-convention re-referencing** (§10, other agent)
   - **Why:** 2026 Savant changed the plate convention (ABS zone, 13% shorter).
   - **Strategy:** quantile-map 2026 plate_z onto 2025.
   - **Result:** 2026 grades move about 1 point on average. Applied in the export.
4. **AAA backfill and AAA vs MLB audit** (§14–15)
   - **Data:** 800 AAA days, 2023–26, via the Savant minors endpoint.
   - **Strategy:** compare the same pitchers at both levels.
   - **Result:**
     - Tracking offsets: extension +0.10 ft, release height –0.06 ft.
     - The fastball-ride gap is altitude; five PCL parks are excluded.
     - CSW% level gap: –1.8 points.
5. **Memorization check** (§17): the most important finding.
   - **Model:** v2 LightGBM.
   - **Strategy:** compare per-pitch fit on pitchers the model trained on vs held-out pitchers, in the same seasons.
   - **Result:** seen .38 vs unseen .055. The model fingerprints pitchers through release point and the fastball-relative inputs. Game folds and season holdouts cannot detect this.
   - **Fix:** the export switched to pitcher cross-fit (owner approved).

B. Target and model bake-offs (earlier work, other agent)

6. **Target bake-off** (§4)
   - **Models:** LightGBM on four targets (CSW-normalized, run value and others).
   - **Result:** `csw_norm` won.
   - **Caveat:** it ran on the memorizing harness, so it is stale. Replaced by #27.
7. **Physics bundle** (§3)
   - **Features:** drag, Magnus-implied ride and run, total Magnus break, active spin.
   - **Result:** rejected as a bundle, also on the old harness. Active spin was later adopted alone.
8. **Mechanism study** (`MECHANISM_MLB.md`)
   - **Method:** descriptive Hawk-Eye analysis.
   - **Result:**

        - Active spin helps fastballs (+1.9) and hurts sliders (–2.2, the gyro effect).
        - Velocity and shape multiply rather than add.
9. **Stuff+ v2** (shipped)
    - **Design:** v1 plus `spin_eff` (monotone per class) plus velo × ivb.
    - **Gates:** passed the location check against v1 (§11).
10. **Game-fold out-of-fold check** (§13)
    - **Result:** YoY unchanged, so stickiness looked real. This check cannot see memorization (#5).

C. Studies on the honest harness

11. **Small-sample regime** (§9, §18)
    - **Question:** first-N-pitch Stuff+ vs CSW% predicting rest of season.
    - **Result:**
        - Stuff+ alone beats results only through about 25–50 pitches.
        - A Stuff+ + results blend is best at every N.
12. **Survivorship with demoted arms** (§15)
    - **Strategy:** fill the rest of season with AAA, minus the level gap.
    - **Result:** the pooled claim holds; the demoted subset is inconsistent (n about 60).
13. **tjStuff+ head-to-head** (§19)
    - **Model:** tjStuff+ LightGBM, rebuilt from its model file (r .966 to published).
    - **Result:** ours wins next CSW% 3/3 and K% 1/3; tjStuff+ is stickier. Neither beats naive.
14. **Spin-axis deviation** (§16, §20)
    - **Feature:** observed minus movement-implied axis, the seam-shifted-wake angle.
    - **Result:** passed on the old harness, failed on the honest one (YoY –.012 to –.016). Rejected.
15. **Bat-tracking swing disruption** (§21)
    - **Data:** 2024–26 bat speed, swing length, contact depth, each relative to the batter's own norm.
    - **Result:** timing disruption is a persistent pitcher trait and adds +.006 / +.009 on CSW% after a pitch-mix control. Contact depth alone is pitch mix.
16. **Reliever leverage next season** (§23)
    - **Method:** empirical gmLI, the leverage teams actually give a reliever.
    - **Result:** Stuff+ adds +.029 / +.024 / +.002 beyond leverage and results. Teams act on stuff.
17. **Measured arm angle** (§24)
    - **Feature:** Hawk-Eye pose arm angle.
    - **Result:** next CSW% better 3/3, significant 2/3. Pass; proposed for v3.
    - **Variant:** arm angle in place of release height and side fails, but cuts the leak by .026.
18. **Empirical-Bayes blend** (§25)
    - **Model:** one frozen shrinkage rule for blending first-N-pitch CSW% with Stuff+.
    - **Result:** broke on the 2026 level shift. Rejected.
19. **Pitchability residual** (§26)
    - **Question:** is CSW% above Stuff+'s expectation persistent?
    - **Result:** YoY about .50, and it survives a pitch-mix control. Chase rate and first-pitch strikes explain a slice.
20. **AAA → MLB translation** (§27)
    - **Model:** AAA pitches graded by the export fold models.
    - **Result:** for prospects with no MLB time that season, AAA Stuff+ beats their AAA K% for next MLB K%: r .26 vs .18 (n 169). Both together is best.

D. Feature selection and importance (§28–29)

21. **Feature importance** (`feature_importance.py`)
    - **Methods:** gain, mean |SHAP|, and permutation loss on seen vs unseen pitchers, overall and by pitch type.
    - **Result:**
        - Gain overstates raw spin and release side; their value is pitcher fingerprinting.
        - The true per-pitch signal is hvaa, release height and ride.
22. **Ablations** (9 cross-fit runs plus bootstrap)
    - **Keep:**
        - Release point.
        - Fastball-relative inputs.
        - Extension, which carries next-season K%.
        - hvaa, which costs fit if removed (but see #25).
    - **Removable:** raw spin. Removing it costs no fit and nudges K% up.
    - **Removable by the rule:** active spin + velo × ivb, though K% leans keep.

- **Rejected:**
  - Drag: YoY –.02, leak +.010.
  - A stripped-down 7-input model: less fit, no gain.

23. **Separate models per Statcast pitch type** (8 types × batter side)
    - **Result:** fit .050 vs .058–.061; next CSW% –.027 / –.044* / –.039*. Rejected.
    - Per-type grades should come from the class models.
24. **Non-Magnus movement** (inches; measured minus Magnus-implied break)
    - **Result:** +.008 fit, the best honest per-pitch gain, with no location channel. YoY –.012 / –.018.
    - Rejected for Stuff+; useful for pitch design.
25. **Horizontal approach angle** (residualized on plate_x)
    - **Result:** fit .137 (doubled) because it rebuilds plate_x (R^2 .91–.999). Rejected.
    - **Also found:** the same geometry means hvaa rebuilds fastball plate height (R^2 .86).
    - **Also found:** the plate_z-only leak gate missed this, so a plate_x leak gate was added.

E. Model-structure program (§30, owner approved)

26. **Settings tuned on unseen pitchers** (`tune_lgbm.py`)
    - **Grid:** num_leaves {7, 15, 31, 63} × min_child_samples {100, 1000, 5000}, early stopping on fold 0.
    - **Winner:** 15 leaves, 1,000 rows per leaf, 71 trees.
    - **Result:**
      - Fit rose (.064 / .067).
      - YoY fell to .72–.74.
      - Plate-height leak rose to .346.
      - Next-season bootstrap all inside noise.
    - **Fails.** Lesson: tuning on per-pitch fit rewards location; tune on next-season pitcher outcomes instead.

27. **Target comparison** (same features and harness; bootstrap vs the CSW target)

Target	Next K%	Next RV/100	Next CSW%	YoY	Leak z / x
Whiff per swing	+.070* / +.092* / +.044	–.017 / +.087* / –.005	–.038 / –.038 / –.082*	.82 / .86 / .80	.161 / .125
Swinging strikes per pitch	+.059 / +.072* / +.028	–.029 / +.067 / –.019	–.033 / –.054 / –.083*	.81 / .86 / .81	.139 / .116
Raw run value	–.047 / –.029 / –.073*	–.055* / –.024 / –.059*	–.061* / –.072* / –.081*	.84 / .87 / .87	.014 / .046
Run value minus count + location	–.029 / –.007 / –.044	–.023 / +.009 / –.041	–.048 / –.043 / –.083*	.83 / .85 / .86	.014 / .027

    - **By the pre-registered rule, no target replaces CSW,** because CSW% gets worse.
    - **Read:** whiff per swing is the best strikeout and run-value target, and it leans less on location. CSW loses because called strikes are location.
    - A Stuff+ (whiff) + Location+ (called strikes) split needs its own pre-registered test.

28. **Additive model** (boosted stumps, max_depth 1; a GAM/EBM-style model with no interactions)
    - **Result:** ties LightGBM on every outcome (all intervals cross zero) and is stickier (.82 / .84 / .82).
    - The algorithm is not the bottleneck.

29. **Production-style decomposed model, v0** (`stuff_decomp.py`)
    - **Design:** PitchingBot-style LightGBM sub-models per class × batter side:
      - swing
      - whiff given a swing
      - foul given contact
      - called strike given a take
      - run value in play
    - **Inputs:** shape + location + count.
    - **Grades:** Stuff+ = expected run value at the league location and count mix (one random draw per pitch). Pitching+ = expected run value at the actual context. Location+ = the difference.
    - **Result:**
      - Leak gone (.006 / .014).
      - Stuff YoY .63–.68.
      - Next K% –.07* and CSW% –.08* in 2025→26.

        - Pitching version: next RV/100 +.101* in 2024→25.
    - **Fails as built.**

30. **Decomposed model v1** (running)

    - **Changes from v0:**
        - The in-play target is a count-specific linear map of Statcast xwOBA onto run value (removes runners, defense and luck).
        - Stuff+ averages 24 shared contexts per pitch (common random numbers) on at most 250 pitches per pitcher-season-cell.
    - **Result:** pending.

F. Not tried yet (ranked by expected gain per day)

1. **Whiff-based Stuff+ with a separate called-strike Location+** (pre-register first). Follows #27.
2. **Double machine learning.**
    - **What:** cross-fit residualize the target AND every shape input on location and count, then fit residual on residual.
    - **Why:** removes hidden location channels (hvaa, HAA) without dropping the inputs.
3. **Tuning on the right objective.** Choose settings by next-season pitcher-level r on held-out pitchers, not per-pitch fit (#26).
4. **v3 inputs on the winning structure:** drop raw spin and hvaa, add arm angle; judge with both leak gates.
5. **Pitcher grades with intervals.** Hierarchical Bayesian aggregation; gives the site its missing standard errors.
6. **Location+ and command.** From the decomposed models; command = best E[RV] location vs actual. Cf. xCTRL.
7. **Arsenal and tunneling.**
    - Distance between consecutive pitches at the decision point (about 24 ft).
    - Pitch-type predictability by count.
    - Mix+ / Match+.
    - Tested against the persistent residual (#19).
8. **Late break:** movement after the decision point, from the 9-parameter trajectory.
9. **Swing-response targets.** Bat speed loss, contact depth and miss distance as multi-task targets beside whiff (follows #15).
10. **Aggregated targets.** Pitch-type season whiff rate from average shape (a cleaner target).
11. **Adversarial neural encoder.** A gradient-reversal location head; only if double ML falls short.
12. **Monotone-constraint ablation.** The constraints have never been priced.
13. **Primary-fastball anchor by usage, using only past pitches.** Removes in-season future leakage.
14. **Rolling retrain.** Add 2025 to training; drift monitors on each new season.
15. **Topo tagger key on MLB.** Requested from the other agent; test it as a display layer or as an input, not as cells (#23).
16. **Transfer to college.** MLB-trained, fine-tuned on D1 TrackMan with measured offsets.
17. **Sequence transformer with player embeddings.** Low priority: high memorization risk, and the college study failed.